



MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

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Chair of Empirical Microeconomics

ADVANCED MICROECONOMICS

Consumer Theory III - Expenditure Minimization

Winter Term 2025/2026

Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Duality (L4+L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Safety Regulations and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

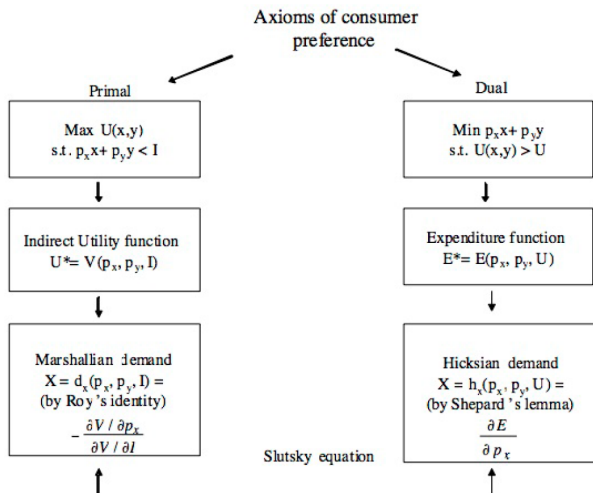
Literature

- Autor (2016): Lecture Notes 5
- **Application:** Waldfogel, J. (1993) “The deadweight loss of Christmas”, American Economic Review, 83(5): 1328–1336.

Overview

- 1 Expenditure function
 - Graphical Illustration
 - Mathematical Solution
- 2 Uses of the expenditure function
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Duality



Source: Autor (2016), Lecture Notes 4, p. 2

Expenditure minimization and utility maximization

- **Expenditure minimization** is the **dual approach** to utility maximization (primal approach) for modeling consumer decisions.
- Similar inverse approaches in situation of firms that decide on input and output that determine profit
 - Approach 1: Profit maximization, choosing input and output for given prices
 - Approach 2: Cost minimization at specific output level
- Utility maximization delivers the **indirect utility function**; expenditure minimization gives an **expenditure function**.
- This is particularly useful as it translates utility levels in a “money metric”, i.e., a cardinal unit that allows comparing different situations.

Overview

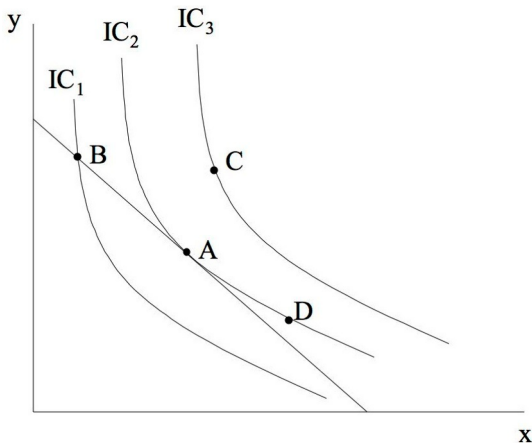
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- 2 Uses of the expenditure function
- 3 The Carte Blanche principle
- 4 Application: The deadweight loss of Christmas
- 5 Summary

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Recap: Utility maximization - graphical illustration

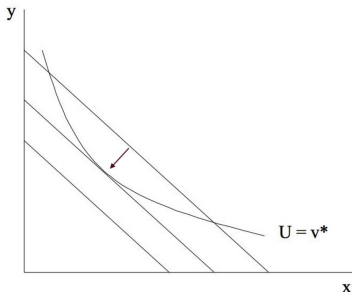
- **Indifference curve** with the **highest possible utility** - Utility maximization in A



Source: Autor (2016), Lecture Notes 4, p. 3

Expenditure minimization - graphical illustration

- **Budget constraint** at which a specific utility level (here v^*) is reached with **minimal possible expenditures**:



Source: Autor (2016), Lecture Notes 5, p. 3

- Both approaches deliver same condition for optimality: **psychic trade-off (MRS) = market trade-off (price ratio)**, and the same optimal bundle of goods

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Approaches

1: Utility maximization (primal approach)

$$\max_{x,y \geq 0} U(x,y) \text{ s.t. } p_x x + p_y y \leq I$$

Solution delivers the optimal amounts x^*, y^* (as functions of p_x, p_y , and I), plugging them into the utility function gives $u^* = U(x^*, y^*)$ for given p_x, p_y, I . In general, we get the **indirect utility function** $V(p_x, p_y, I) = u^*$.

2: Expenditure minimization (*cost minimization*) (dual approach)

$$\min_{x,y \geq 0} p_x x + p_y y \text{ s.t. } U(x,y) \geq u^*$$

Solution delivers optimal amounts x^*, y^* (as functions of p_x, p_y , and u^*), plugging into the budget restriction (*cost function*) gives the **expenditure function**: $E(p_x, p_y, u^*) = p_x x^* + p_y y^*$.

An example

continued from Lecture 2

- **Example** $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$
- The primal approach delivers for every combination of prices, p_x, p_y , and income I the maximum U_p
- The **expenditure minimization approach** is

$$\min_{x, y \geq 0} p_x x + p_y y \text{ s.t. } x^{\frac{1}{2}}y^{\frac{1}{2}} \geq U_p$$

- Solving this requires again the **Lagrange-approach**

$$\mathcal{L} = p_x x + p_y y + \lambda(U_p - x^{\frac{1}{2}}y^{\frac{1}{2}})$$

An example - continued

First Order Conditions (FOCs)

$$\textcircled{1} \quad \frac{\partial \mathcal{L}}{\partial x} = p_x - \lambda \frac{1}{2} x^{-\frac{1}{2}} y^{\frac{1}{2}} = 0$$

$$\textcircled{2} \quad \frac{\partial \mathcal{L}}{\partial y} = p_y - \lambda \frac{1}{2} x^{\frac{1}{2}} y^{-\frac{1}{2}} = 0$$

$$\textcircled{3} \quad \frac{\partial \mathcal{L}}{\partial \lambda} = U_p - x^{\frac{1}{2}} y^{\frac{1}{2}} = 0$$

Solving FOCs 1 and 2 for λ and rearranging yields

$$x^* = \frac{p_y y^*}{p_x}$$

An example - continued

Plugging into FOC (3) yields $U_p = \left(\frac{p_y y^*}{p_x}\right)^{\frac{1}{2}} (y^*)^{\frac{1}{2}} = \left(\frac{p_y}{p_x}\right)^{\frac{1}{2}} y^*$
and thus

$$y^* = \left(\frac{p_x}{p_y}\right)^{\frac{1}{2}} U_p \text{ and } x^* = \left(\frac{p_y}{p_x}\right)^{\frac{1}{2}} U_p.$$

Plugging into budget restriction gives **expenditure function**

$$E(p_x, p_y, U_p) = p_x \left(\frac{p_y}{p_x}\right)^{\frac{1}{2}} U_p + p_y \left(\frac{p_x}{p_y}\right)^{\frac{1}{2}} U_p = 2p_x^{\frac{1}{2}} p_y^{\frac{1}{2}} U_p.$$

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Why do we need the expenditure function?

- Recap: Indirect utility function delivers maximal possible utility but its value cannot (without further assumptions) be interpreted cardinally (only ordinal).
- As we cannot interpret the utility function cardinally, it's difficult to use it to evaluate impacts of policy interventions
 - Before a policy intervention a household reaches U_0
 - After the policy intervention the household reaches $U_1 > U_0$, i.e., the household is better off.
 - But by how much??
- This question is particularly important if we want to find out how much households value different measures. (How much would they be willing to pay for it?)
- The expenditure function helps because it allows translating utility levels into costs, i.e., a money metric, which can be interpreted cardinally.

Compensation for a policy measure

- **Compensation:** The expenditure function allows for calculating how much money a household would have to receive to be compensated for a specific policy intervention, i.e., to keep utility level constant.
 - The compensation payment can also be negative if the change has a positive impact on the household (e.g. decreases in child care costs, decreases in Mensa prices, decreases in gas prices).
- There are different concepts for calculating compensation payments:
 - 1 Compensating variation (CV)
 - 2 Equivalent variation (EV)

Compensating variation (CV)

- **Example** Increase in mensa prices from s_0 to s_1
 - Known: U_0 maximal utility before increase, p_a all other prices
- **Compensating variation (CV)** is defined as

$$CV = E(s_0, p_a, U_0) - E(s_1, p_a, U_0)$$

- CV is amount of money that at – constant prices of all other goods p_a – is necessary to keep U_0 despite the increase in s
 - Or: CV measures, how much a person would have to receive as a payment **after a change** in prices (as a compensation) to still reach the pre-change utility level.
- This concept is often used in **cost-benefit-analyses** to evaluate policy measures.

Equivalent variation (EV)

- **Example** Increase in mensa prices from s_0 to s_1
 - Let U_i denote the maximum utility reached with s_i
- **Equivalent variation (EV)** is defined as

$$EV = E(s_0, p_a, U_1) - E(s_0, p_a, U_0)$$

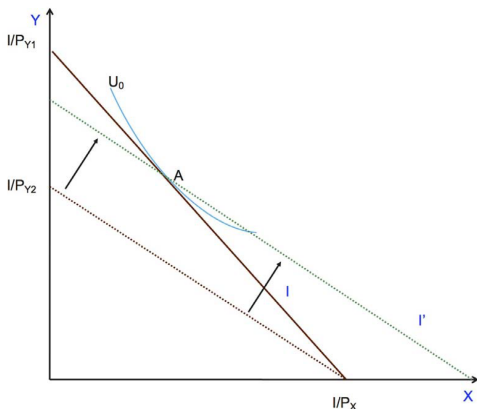
- EV measures how much the change in utility ($U_1 - U_0$), that results from the price change (s_1 instead of s_0), is worth in a money metric
 - EV thus can be interpreted as the amount a person would be willing to pay (maximally) **ex ante** (i.e., with p_a and s_0) to prevent an increase to s_1
- This concept is often used to measure the **welfare loss** (deadweight loss) of a policy

Compensations in reality

- To calculate compensation payments (CV or EV) in reality, one needs (to know or to assume) a utility function, which is unrealistic
- Alternatively one could offer as a compensation the amount of money that the household would need to **buy the same bundle of goods** as before a price change: $\Delta s \times Q_s$, where $\Delta s = s_1 - s_0$ and Q_s amount of mensa food bought before price change.
- This compensation would allow the household to reach a higher utility level than before the price change →

Overcompensation

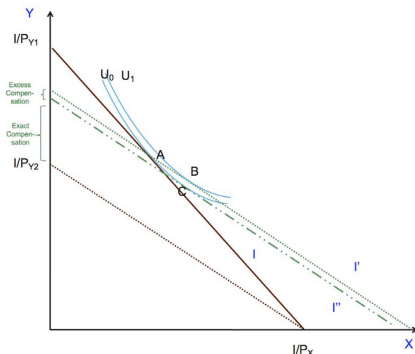
Overcompensation - Graphical Illustration



Source: Autor (2016), Lecture Notes 5, p. 7

- Y good of interest, P_Y price of Y , X all other goods with price P_X , I income
- A bundle of goods consumed before change in price of Y .
- U_0 utility level before price change.
- P_Y increases from P_{Y1} to P_{Y2} .
- Compensation to I' , so that same bundle can be bought.

Overcompensation and exact compensation



Source: Autor (2016), Lecture Notes 5, p. 8

- **Exact Compensation (to I'')**

$$CV = C^* = E(P_{Y2}, P_X, V(P_{Y1}, P_X, I)) - E(P_{Y1}, P_X, V(P_{Y1}, P_X, I))$$

- **BUT:** But A is not the optimal bundle given I' .
- The optimal bundle would be B , which delivers a higher utility than A ($U_1 > U_0$) $\rightarrow$ i.e., change to I' overcompensates for the price change.
- For an exact compensation the household does not need to reach A but IC with same utility as A .
- Compensation to I'' would suffice.

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Policy Implication: The Carte Blanche principle

- **Standard assumption:** Households always chose the consumption bundle that makes them as well off as possible based on given prices and income
- Consumers cannot be made worse off by a **free monetary transfer** (i.e., an unrestricted cash transfer or payment) compared to **in-kind transfers** (or restricted transfers).
- Why?

Carte Blanche Principle

Households will always be weakly better off with a **cash transfer** than with any other transfer.

- Cash transfers give a *Carte Blanche* to buy whatever one likes with the money (including the in-kind transfer that would be given)

Example: Government Intervention

- Assumption: Households spend money on food f and housing h , with utility function $u(f, h)$ and prices p_f and p_h , respectively
- Without government intervention households have an income I , which they can divide between f and h , leading to budget constraint: $I \geq p_f f + p_h h$.
- **Introduction of subsidized housing for the poor**, with a bonus payment for housing of 50 €
 - **Two budget constraints:** $I + 50 \geq p_f f + p_h h$ and $I \geq p_f f$.
 - If the government just handed out a monetary transfer of 50, households would only face one constraint:

$$I + 50 \geq p_f f + p_h h.$$
- **Standard micro theory:** Households cannot be better off with more restrictions.

Example: Government Intervention

- There are, however, many situations in which governments/NGOs rely on **in-kind transfers** or **conditional cash transfers** rather than pure cash transfers
 - Examples: Housing vouchers, food stamps, conditional cash transfers (esp. in developing countries)
- **Why?** - Worry that people are not able to make decisions in their own best interest (**paternalistic view**) or demand of funders (e.g. tax payers)
- **Empirical Evidence:** Different studies show that unconditional cash transfers can often reach goals as well as conditional cash or in-kind transfers (e.g., Cunha, 2014 or study on use of child allowance in Germany Stichnoth et al., 2018)

Further Resources

Empirical Studies

- Cunha, J.M. (2014) “Testing Paternalism: Cash versus In-kind transfers” American Economic Journal: Applied Economics, 6(2): 195–230
- Haushofer, J. and J. Shapiro (2016) “The short-term impact of unconditional cash transfers to the poor: experimental evidence from Kenya” Quarterly Journal of Economics, 131(4): 1973–2042
- Stichnoth, H. et al. (2018) “Kommt das Geld bei den Kindern an?” Bertelsmann Stiftung.

Podcasts on conditional cash transfers and cash vs. in-kind transfers

- “Episode 665: The Pickle Problem.” Planet Money the Economy Explained, November 25, 2015.
<https://www.npr.org/player/embed/457429587/457430507>
- “Episode 494: What Happens When You Just Give Money To Poor People?” Planet Money the Economy Explained, November 8, 2013
<https://www.npr.org/sections/money/2013/11/08/243967328/episode-494-what-happens-when-you-just-give-money-to-poor-people>
- AEA Research Highlights Podcast, Ep. 18: The promise of conditional cash transfers, Dezember 23, 2020 <https://aearesearchhighlights.libsyn.com/ep-18-the-promise-of-conditional-cash-transfers>
- Why Is This Happening? The Chris Hayes Podcast, The Case for Unconditional Cash Transfers with Jiaying Zhao, Dezember 31, 2024

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The Deadweight Loss of Christmas

Waldfoegel, 1993, American Economic Review 83(5)

The Deadweight Loss of Christmas

By JOEL WALDFOGEL*

When economists comment on holiday gift-giving, it is usually to condone the healthy effect of spending on the macroeconomy. However, an important feature of gift-giving is that consumption choices are made by someone other than the final consumer. A potentially important *microeconomic* aspect of gift-giving is that gifts may be mismatched with the recipients' preferences. In the standard microeconomic framework of consumer choice, the best a gift-giver can do with, say, \$10 is to duplicate the choice that the recipient would have made. While it is possible for a giver to choose a gift which the recipient ultimately values above its price—for example, if the recipient is not perfectly informed—it is more likely that the gift will leave the recipient worse off than if she had made her own

representative of the U.S. population, their *gifts* are not necessarily unrepresentative. Holiday expenditures average \$40 billion per year, implying that a conservative estimate of the deadweight loss of Christmas² is a tenth as large as estimates of the deadweight loss of income taxation. I also explore how deadweight loss and the tendency to give cash gifts vary with the relationship and age difference between giver and recipient. I find that gifts from friends and “significant others” are most efficient, while noncash gifts from members of the extended family are least efficient and destroy a third of their value. I develop a simple expected-utility model to explain the decision to give cash, as opposed to in-kind gifts. The data are consistent with the model: cash gifts are most common for distant

Questions about the study

- What is the paper about?
- How is it related to our class?
- What data and methods are used?
- What are the paper's (main) results?
- How valid are these results?
- What is your personal take-away from this paper?

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Summary

- Cost minimization dual approach to utility maximization.
- The expenditure function is particularly useful as it translates utility levels into a money metric.
- It can be used to measure the value of policy measures (compensation payments, as CV or EV).
- According to standard theory cash (weakly) better than in-kind transfers and gifts.
- In-kind transfers still common (although evidence suggests that doubts about usefulness of cash transfers not always correct).
- Further illustration by Waldfogel's "deadweight loss of Christmas".

Main Keywords

- ➊ Duality of utility maximization and expenditure minimization
- ➋ Expenditure function $E(p_x, p_y, U)$
- ➌ Compensating variation (CV)
- ➍ Equivalent variation (EV)
- ➎ Compensation payments
- ➏ Carte blanche principle
- ➐ Cash vs. in-kind transfers
- ➑ Conditional cash transfers / tied transfers

Questions?

